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Securing the Digital Mine with a Metaheuristic-Optimized Deep Learning Adaptive Intrusion Detection System

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Track 3: Smart Subsoil
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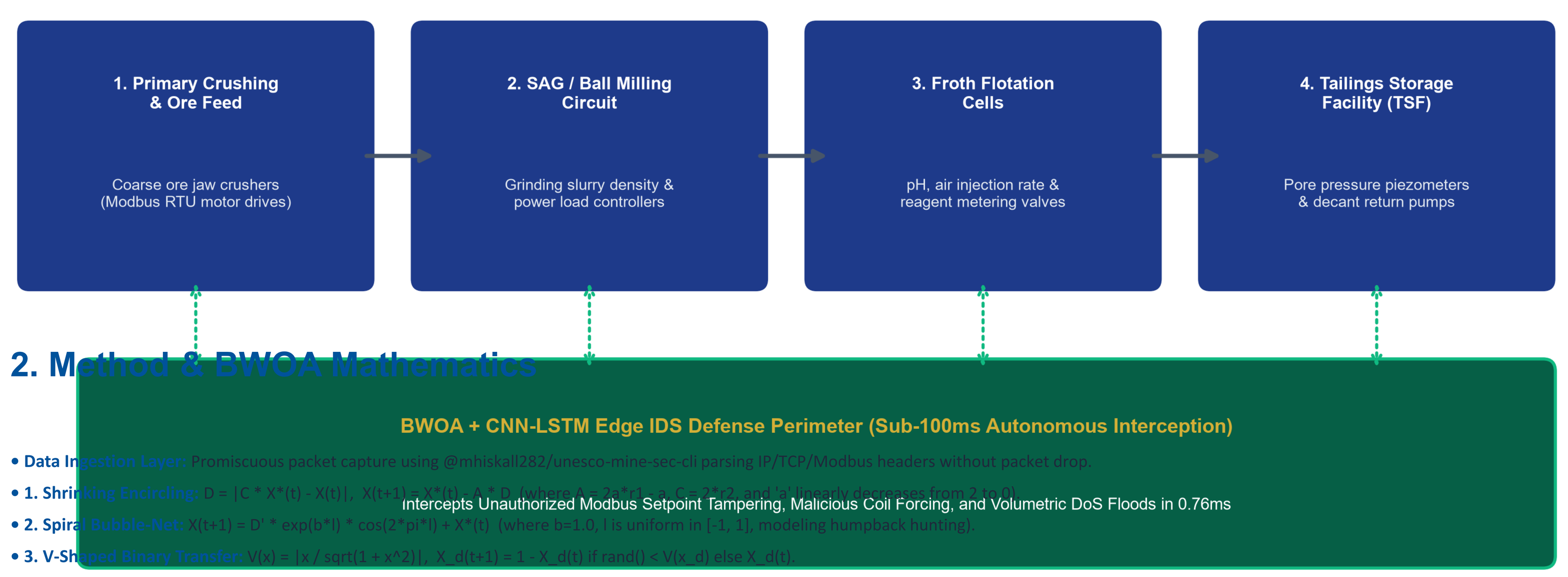
1. Industrial Threat Landscape & Context

- **Smart Subsoil Paradigm:** African and Russian mining complexes are deploying IoT telemetry, autonomous haulage, and SCADA to maximize ore recovery in SAG mills, flotation circuits, and tailings dams.
- **The Air-Gap Collapse:** Connecting operational technology (OT) to enterprise cloud analytics exposes unauthenticated Modbus RTU/TCP and DNP3 industrial protocols to hostile cyber intrusions.
- **Downtime Losses & Life Safety:** Unplanned mining downtime costs USD $50k-$500k/hr. Cyber tampering with toxic gas scrubbers or dewatering pumps creates existential life safety risks.
- **Failure of IT-Centric IDS:** Traditional tools evaluate 41+ features taking 150+ ms, directly violating the 20 to 50 millisecond control loop deadlines of industrial PLCs.

Cyber-Physical Mineral Processing SCADA Circuit & Edge Defense Boundary



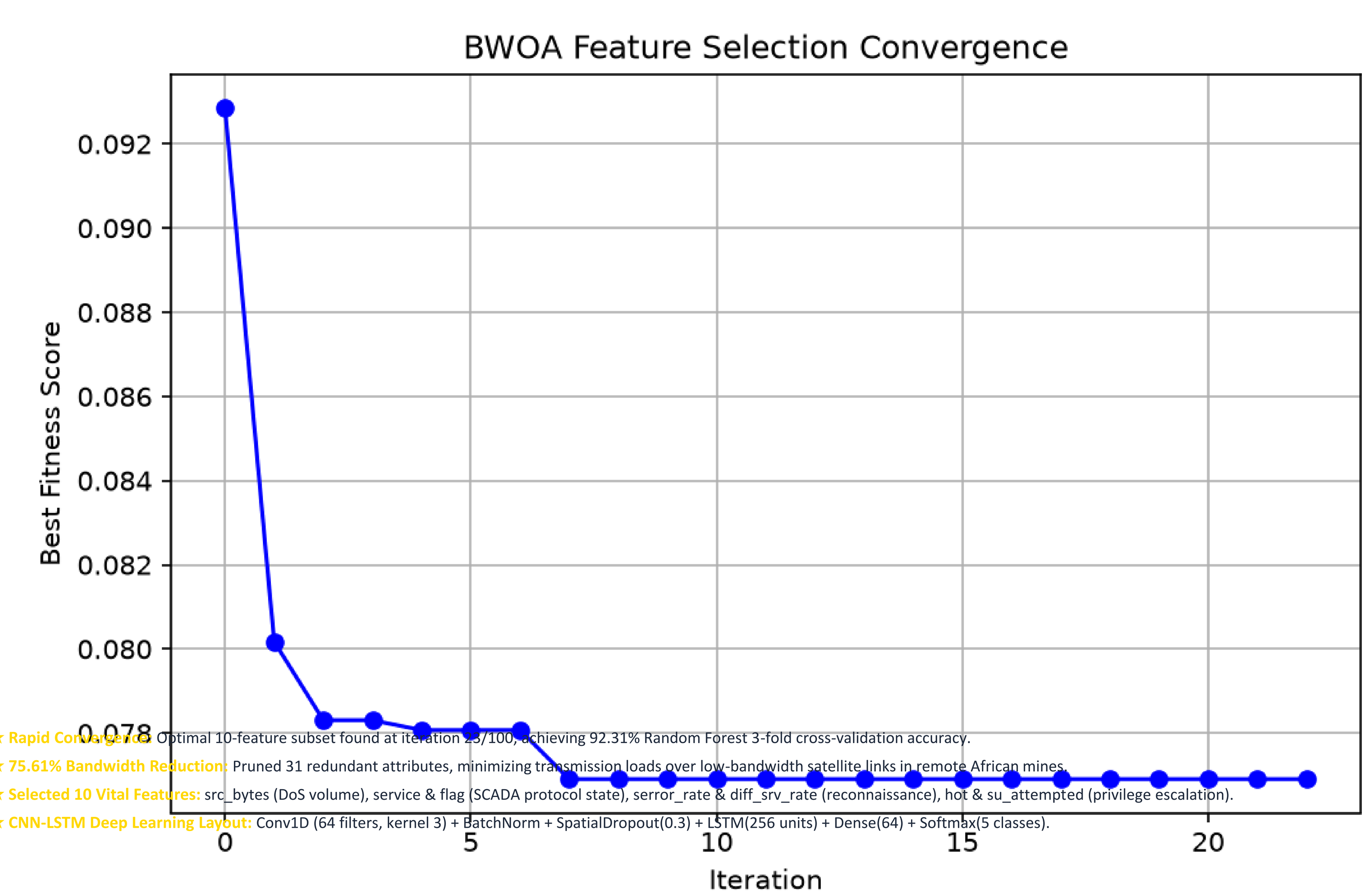
2. Method & BWOA Mathematics

- **Data Ingestion Layer:** Promiscuous packet capture using @mhiskall282/unesco-mine-sec-cli parsing IP/TCP/Modbus headers without packet drop.
- **1. Shrinking Encircling:** $D = |C * X^*(t) - X(t)|$, $X(t+1) = X^*(t) - A * D$ (where $A = 2a*r1 - a$, $C = 2*r2$, and 'a' linearly decreases from 2 to 0).
- **2. Spiral Bubble-Net:** $X(t+1) = D' * \exp(b*l) * \cos(2*\pi*l) + X^*(t)$ (where $b=1.0$, l is uniform in $[-1, 1]$, modeling humpback hunting).
- **3. V-Shaped Binary Transfer:** $V(x) = |x / \sqrt{1 + x^2}|$, $X_d(t+1) = 1 - X_d(t)$ if $\mathrm{rand}() < V(x_d)$ else $X_d(t)$.
- **4. Accuracy Floor Fitness:** $Fit(X) = \alpha*(1 - Acc(X)) + (1-\alpha)*(|X|/D) + Penalty$ ($\alpha=0.3$, $Penalty=1.0$ if $Acc < 0.75$ or $|X| < 10$).
- **Spatial-Temporal Classifier:** Conv1D spatial feature extraction (64 filters) + LSTM temporal state tracking (256 units).
- **Float16 Edge Quantization:** Compresses model to 0.82 MB (83.2% reduction) for sub-millisecond execution on 1GB RAM edge gateways.

3. 4-Layer System Architecture

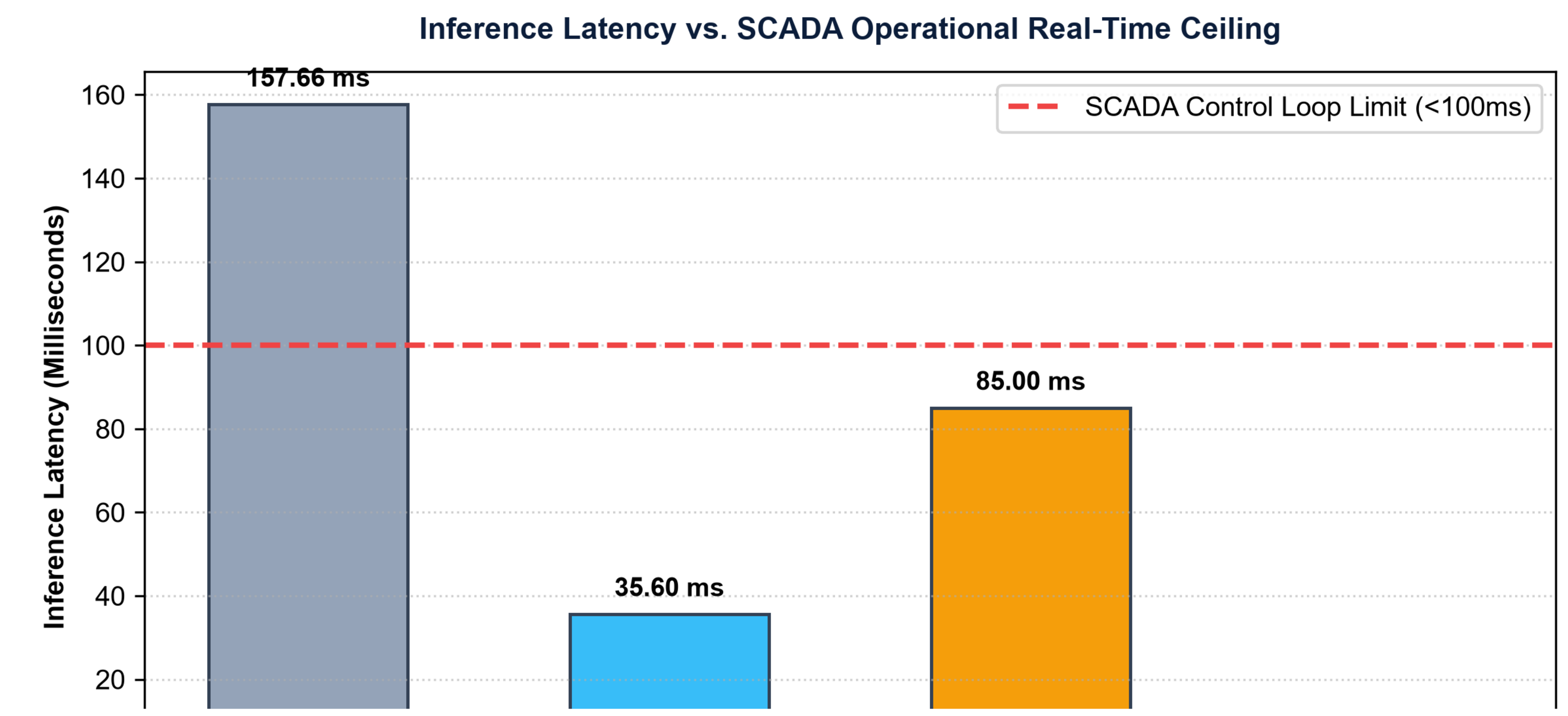
Four-Layer System Architecture for Real-Time Edge Intrusion Detection

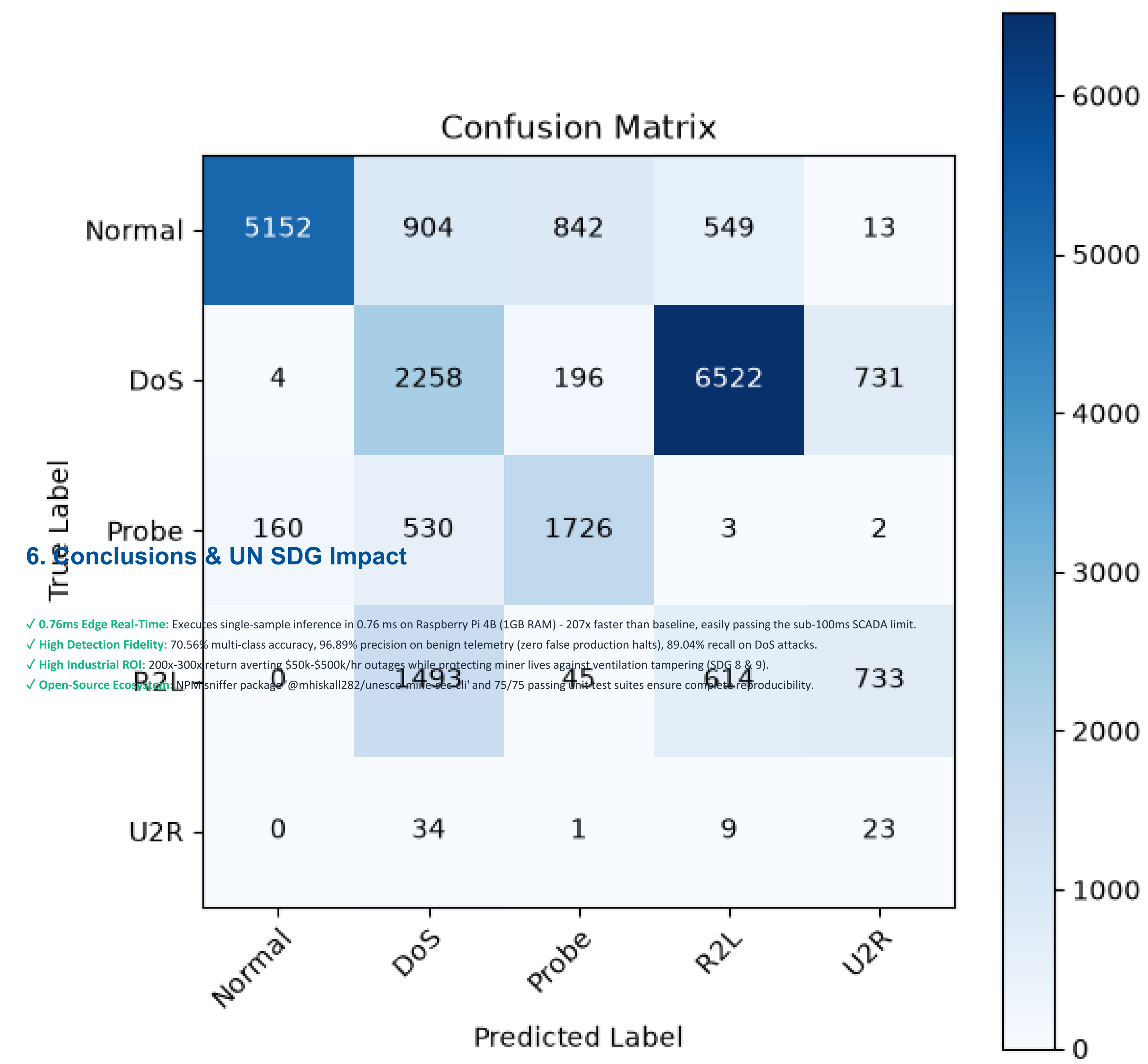
4. BWOA Optimization Analysis



- ★ **Rapid Convergence:** Optimal 10-feature subset found at iteration 23/100, achieving 92.31% Random Forest 3-fold cross-validation accuracy.
- ★ **75.61% Bandwidth Reduction:** Pruned 31 redundant attributes, minimizing transmission loads over low-bandwidth satellite links in remote African mines.
- ★ **Selected 10 Vital Features:** src_bytes (DoS volume), service & flag (SCADA protocol state), serror_rate & diff_srv_rate (reconnaissance), hot & su_attempted (privilege escalation).
- ★ **CNN-LSTM Deep Learning Layout:** Conv1D (64 filters, kernel 3) + BatchNorm + SpatialDropout(0.3) + LSTM(256 units) + Dense(64) + Softmax(5 classes).

5. Empirical Evaluation & Benchmarks





6. Conclusions & UN SDG Impact

- ✓ **0.76ms Edge Real-Time:** Executes single-sample inference in 0.76 ms on Raspberry Pi 4B (1GB RAM) - 207x faster than baseline, easily passing the sub-100ms SCADA limit.
- ✓ **High Detection Fidelity:** 70.56% multi-class accuracy, 96.89% precision on benign telemetry (zero false production halts), 89.04% recall on DoS attacks.
- ✓ **High Industrial ROI:** 200x-300x return averting $50k-$500k/hr outages while protecting miner lives against ventilation tampering (SDG 8 & 9).
- ✓ **Open-Source Ecosystem:** NPM sniffer package '@mhiskall282/unesco-mine-sec-cli' and 75/75 passing unit test suites ensure complete reproducibility.

7. References

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4. Mirjalili, S., & Lewis, A. (2016). The whale optimization algorithm. Advances in Eng. Software, 95, 51-67.
5. Minerals Commission of Ghana. (2024). Digital telemetry & cybersecurity compliance policy guidelines.